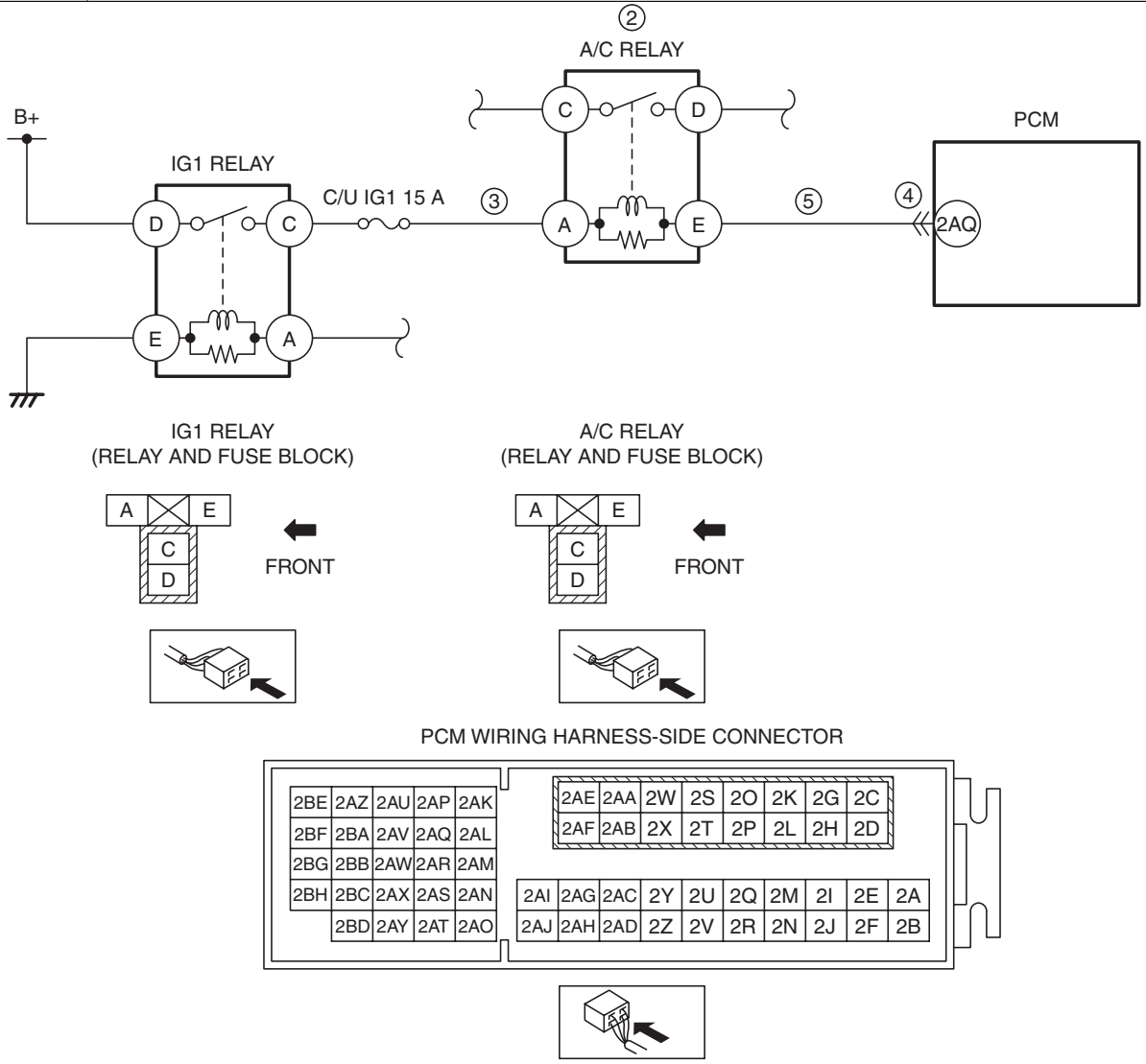
DTC P0647:00 [SKYACTIV-D 1.5]

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DTC P0647:00	A/C relay circuit high input
DETECTION CONDITION	• The current value flowing to the PCM 2AQ terminal is **1 A or more** for a continuous **5 s** when any of the following conditions is met. **MONITORING CONDITIONS** — Battery voltage: **9—16 V** — A/C relay controlled on **Diagnostic support note** • This is an intermittent monitor (other). • The check engine light does not illuminate. • FREEZE FRAME DATA/Snapshot data is not available. • DTC is not stored in the PCM memory.
FAIL-SAFE FUNCTION	• PCM restricts engine-transaxle integration control. (ATX)
POSSIBLE CAUSE	• A/C relay malfunction • Short to power supply in wiring harness between IG1 relay terminal C and A/C relay terminal A • PCM connector or terminals malfunction • Short to power supply in wiring harness between A/C relay terminal E and PCM terminal 2AQ • PCM malfunction



Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
2	**INSPECT A/C RELAY** • Switch the ignition off. • Remove the A/C relay. (See RELAY LOCATION.) • Inspect the A/C relay. (See RELAY INSPECTION.) • Is there any malfunction?	Yes	Replace the A/C relay, then go to Step 6. (See RELAY LOCATION.)
		No	Go to the next step.
3	**INSPECT A/C RELAY POWER SUPPLY CIRCUIT FOR SHORT TO POWER SUPPLY** • Verify that the A/C relay is removed. • Remove the IG1 relay. (See RELAY LOCATION.) • Switch the ignition ON (engine off). **Note** • Another DTC may be stored by the PCM detecting an open circuit. • Measure the voltage at the A/C relay terminal A (wiring harness-side). • Is the voltage **0 V**?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between IG1 relay terminal C and A/C relay terminal A. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to power supply. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to power supply. Go to Step 6.
4	**INSPECT PCM CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the PCM connector. • Inspect for poor connection (such as damaged/pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 6.
		No	Go to the next step.
5	**INSPECT A/C RELAY CONTROL CIRCUIT FOR SHORT TO POWER SUPPLY** • Verify that the A/C relay and IG1 relay are removed. • Verify that the PCM connector is disconnected. • Switch the ignition ON (engine off). **Note** • Another DTC may be stored by the PCM detecting an open circuit. • Measure the voltage at the A/C relay terminal E (wiring harness-side). • Is the voltage **0 V**?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between A/C relay terminal E and PCM terminal 2AQ. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to power supply. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to power supply. Go to the next step.
6	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Start the engine. • Turn the air conditioner on and off. • Perform the DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.

STEP	INSPECTION		ACTION
7	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.